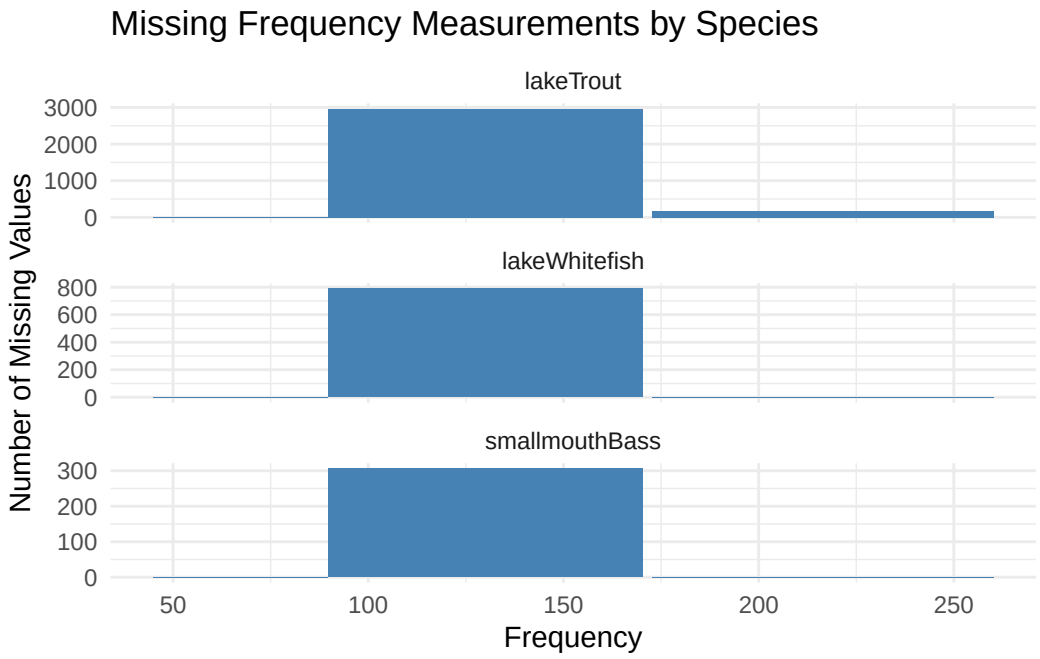
Fish Data: Diagnostics, Missing Values and Correlations

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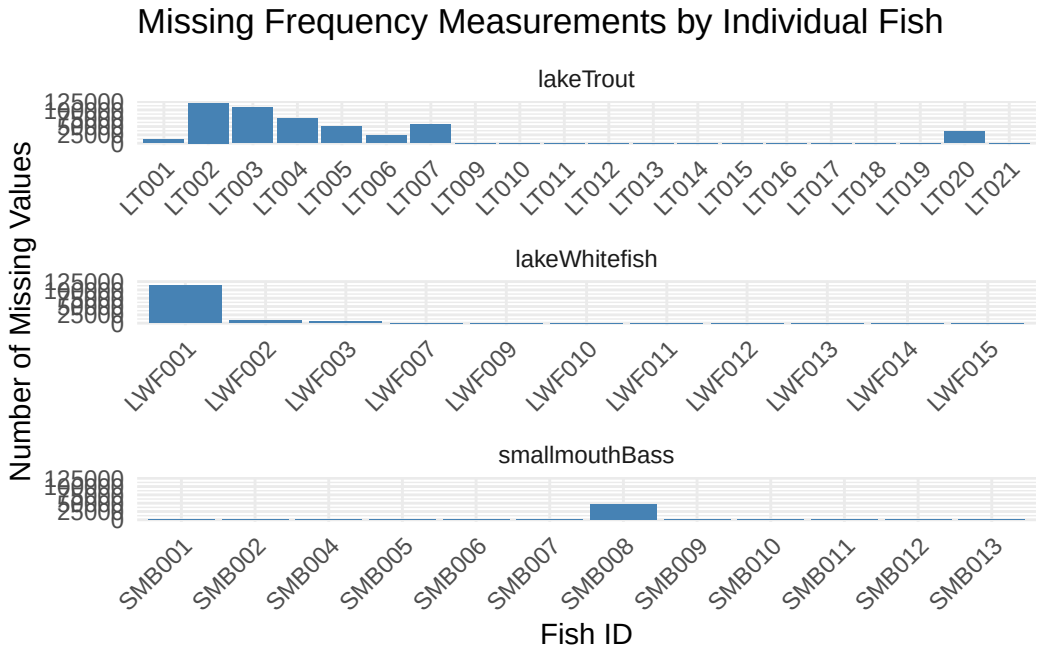
1 Introduction

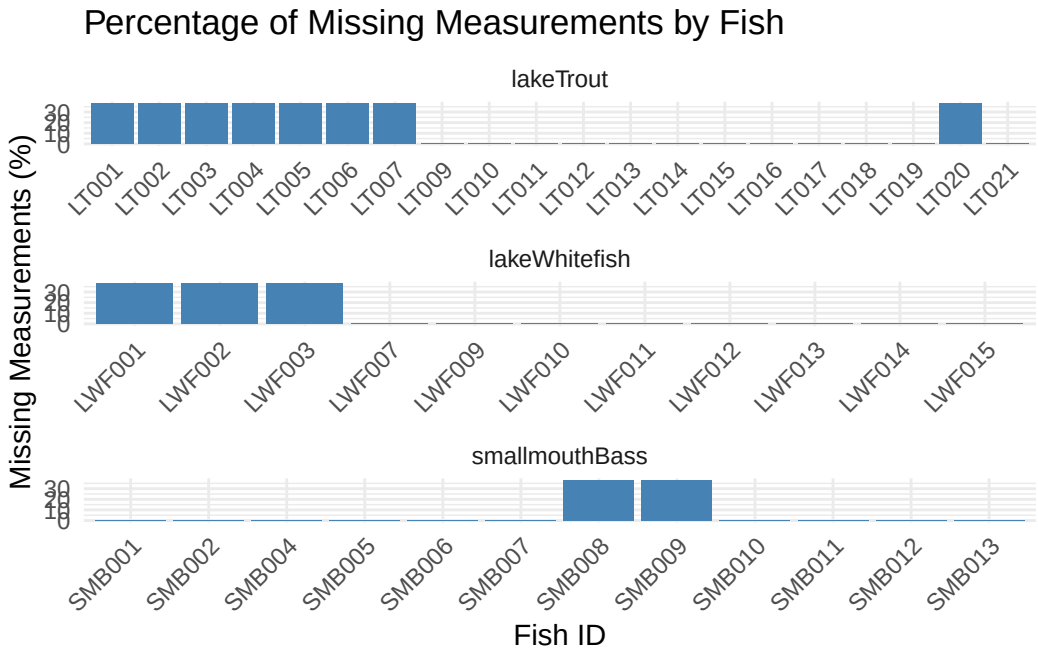
2 Dealing with missing values



Notice the pattern in Figure 1, there are two ranges of frequencies with missing values: F90-F170 and F173-F260. Frequencies F90-F170 are missing for all three species, for almost 3000 observations for LT, 800 for LW and 300 for SMB. Lastly, there are 183 observations in LT with missing values. As shown in Figure 2, there is at least one missing value in 9 LT, 3

LW and 2 SMB. The fish LT008 is the only one missing values between F173-F260, hence we remove it from the dataset.





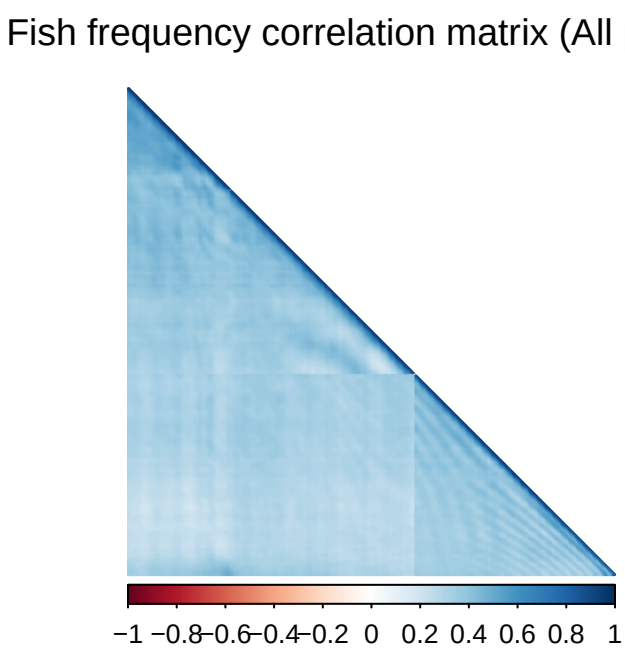
Some fish may be missing more values simply because they have more fish as shown by Figure 2, so we observe Figure 3 which shows percentage of values missing for each fish to account for

this. Among fish with missing observations, approximately 35% of frequency measurements are consistently missing, primarily within the F90–F170 range. This suggests a systematic pattern of missingness rather than isolated missing values. While some imputation methods were considered to replace missing values, given the small number of fish in the sample, the consequences of imputing missing values are unknown, and specially when focusing on dimensionality reduction we want to avoid introducing more noise than there already is to perceive signals clearly. Consequently, we drop any rows that have missing values in the frequencies (4,242 in total).

Table 1: Overview of the dataset with no missing values

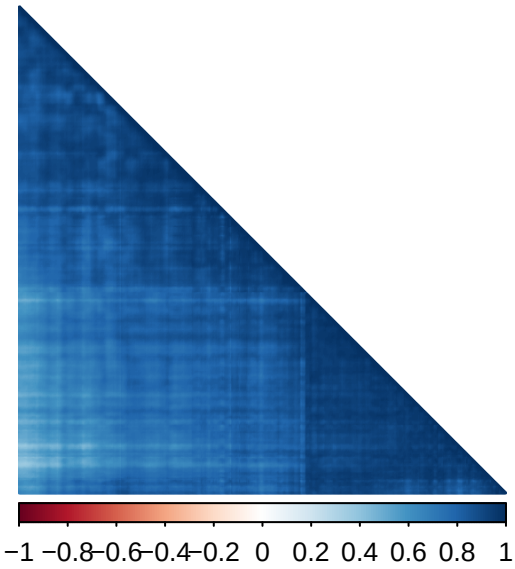
species	Number of fish	Total pings	Median pings	Min. pings	Max. pings
lakeTrout	12	4101	189.5	16	1379
lakeWhitefish	8	3496	312.0	96	1619
smallmouthBass	10	2736	210.0	25	1071

2.1 Exploration

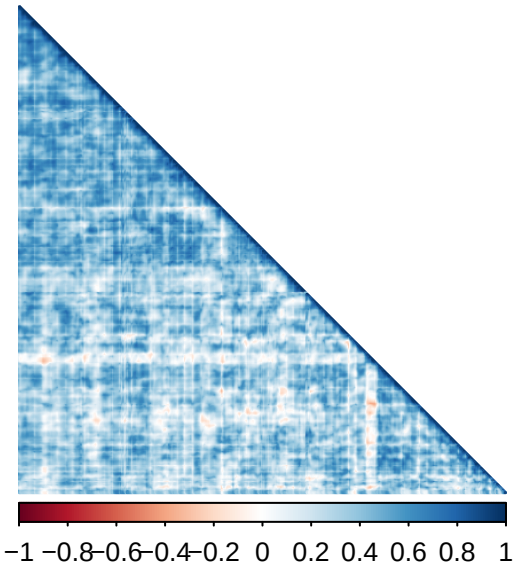


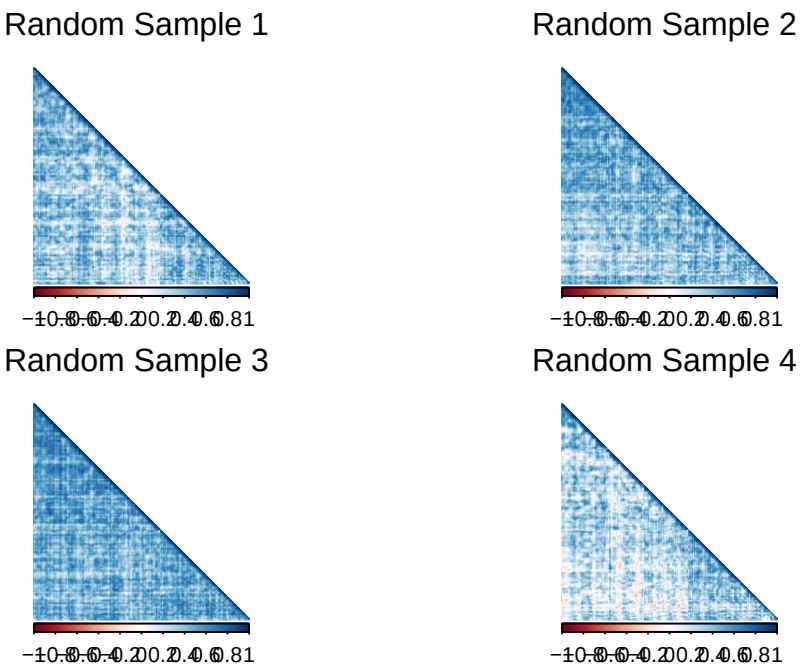
Since each fish contributes multiple pings, correlations in Figure 4 describe relationships across all pings, not across independent fish. Next, we average all frequencies by fish, to obtain one row per fish and plot the correlation matrix again.

Correlation Matrix (One Average Measurement per Fish)

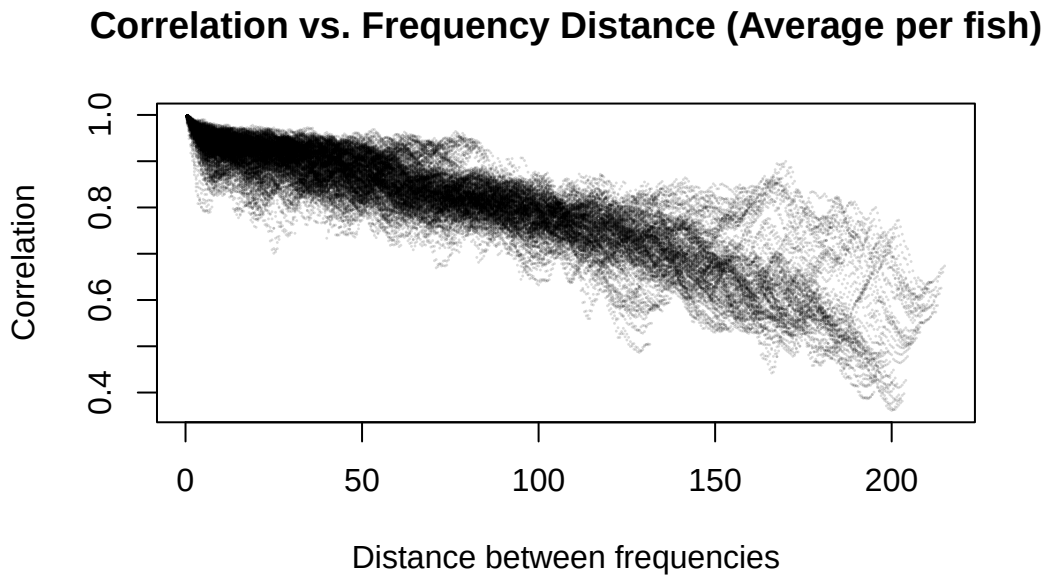


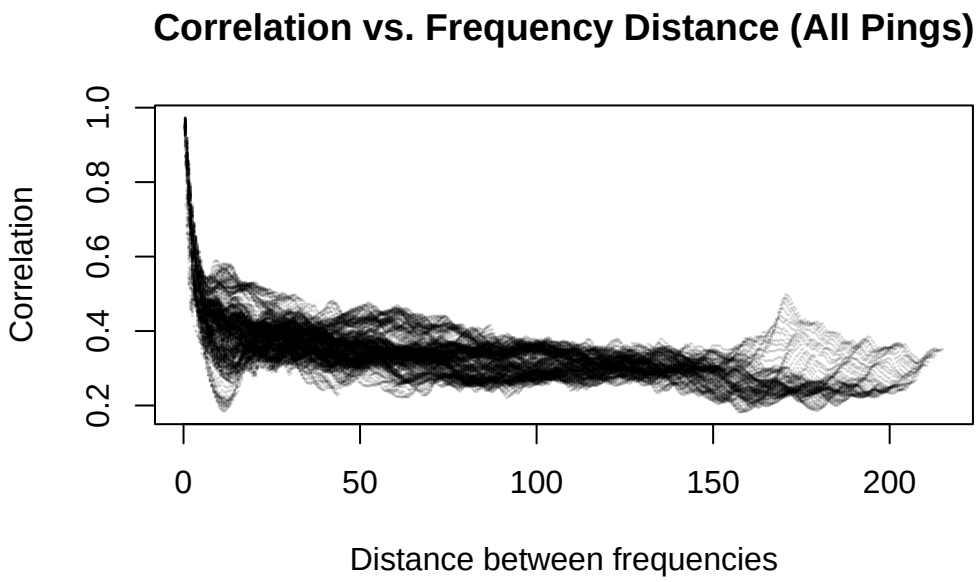
Correlation Matrix (One Random Ping per Fish)





2.2 Closer frequencies are highly correlated




Correlation vs. Frequency Distance (All Pings)
Correlation
0.2
0.4
0.6
0.8
1.0
0
50
100
150
200
Distance between frequencies